

1-Mcq 5 questions: 5 mark

1- The category of computers used in medical electronics (like pacemakers, defibrillators, drug: releasing pumps, hearing aids, etc.) are examples of

- a- Embedded Systems
- b- supercomputers
- c- PMDs
- d- HPC

2- Which drive would the largest storage among those given below

- a) 80 Terabyte
- b) 1 Exabyte
- c) 20 Yottabyte
- d) 3 Zettabyte

3- _____ is an example of system software.

- a) MS Word
- b) Compilers and assemblers
- c) Browsers (Safari, Chrome, etc.)
- d) HD multi-user games

4- A wafer manufactured using 16nm technology used to manufacture 320 dies had a 97.5% yield. What is the number of damaged dies in this case?

- a) $(320) \cdot (97.5/100) \cdot 0.16$
- b) $(16 \cdot 320) \cdot (97.5/100)$
- c) 8
- d) 312

5- Program performance depends on all of these except

- a) Effective Algorithms
 - b) Fast I/O operations
 - c) Efficient IDEs and editors
 - d) Good compilers and assemblers
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2. Translate the following C code to binary code 10 mark

السؤال ذا مهم جدا ودايم الطلاب ينسون انهم يحولون كود الاسبلي الى باينري ركزوا

$$C[7] = (b/4) + a - (8*d)$$

Assume that the variables are stored in the following registers - a: \$s0, b: \$s1 and d: \$s3 and the base address of C is stored in \$s2.

هنا ي جيبون لكم البوكس انها توضح التايب حق الريجيستر بيكون ذا نهايه السؤال والمطلوب منكم

1- تحولون من كود الى كود اسمبلي

2- تحولون من كود اسمبلي الى باينري



3- Consider the C++ code below

```
if ( x1 >= x2 )
    x1++;
else
    x2--;
y1 = ( x1 & x2 ); // does a bitwise AND
```

Assume that x1, x2, and y1 are stored in \$s0, \$s1 and \$t0 respectively <<مهمه<<

a) Translate the above code into assembly code using pseudo instructions. 4 mark

b) Translate the code into standard MIPS assembly code (without using pseudo instructions) 4 mark

Given the initial values of registers \$s0, \$s1, and \$t0, find the register values at the end of execution of the MIPS instructions used in 2 (b) above. (2 marks)

Register Name	\$s0	\$s1	\$t0
Register value (initial)	3	8	4
Register value(final)			



4. Write a procedure in MIPS code that returns the result of the following operation based on the given (single) input parameter. (15 marks)

$$f(x) = x + (x - 1) + (x - 2) + (x - 3) + \dots 1$$

ملاحظة: كان قبل الثلاث نقاط يلي قبل رقم 1 كان في 3 نقاط على فوق مدري اذا تفرق
اول لا نفس ... بس على فوق وذا تقريباً شكله لأنه ما لقيت نقطه على فوق :

1 ... "

5. Suppose you have a program that consists of about 5×10^6 lines of code (LOC). Most of the programs of this type have the following characteristics: (10 marks)

- 30% of the instructions are load instructions (LW) that each take 5 ns to execute.
- 20% of the instructions are store instructions (SW) that each take 4 ns to execute.
- 50% of the instructions are arithmetic and logic instructions (e.g., ADD, SUB) that each take 3 ns to execute.

If this program is executed on a processor with 2.6GHz, calculate the total execution time required for this program.



6. Provide the corresponding assembly language instructions for the two machine codes given below (in hexadecimal) [maybe it will change to from octal or binary](#) 10 mark

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2509001B



- \$a0 - \$a3 Register number 4-7
- \$t0 - \$t7 : Register number 8-15
- \$s0 - \$s7 : Register number 16-23

Name	Format	Example						Comments
add	R	0	18	19	17	0	32	add \$s1,\$s2,\$s3
sub	R	0	18	19	17	0	34	sub \$s1,\$s2,\$s3
addi	I	8	18	17	100			addi \$s1,\$s2,100
lw	I	35	18	17	100			lw \$s1,100(\$s2)
sw	I	43	18	17	100			sw \$s1,100(\$s2)
Field size		6 bits	5 bits	5 bits	5 bits	5 bits	6 bits	All MIPS instructions are 32 bits long
R-format	R	op	rs	rt	rd	shamt	funct	Arithmetic instruction format
I-format	I	op	rs	rt	address			Data transfer format

op(31:26)								
28-26	0(000)	1(001)	2(010)	3(011)	4(100)	5(101)	6(110)	7(111)
31-29								
0(000)	R-format	Bltz/gez	jump	jump & link	branch eq	branch ne	blez	bgtz
1(001)	add immediate	addiu	set less than imm.	set less than imm. unsigned	andi	ori	xori	load upper immediate
2(010)	TLB	FlPt						
3(011)								
4(100)	load byte	load half	lwl	load word	load byte unsigned	load half unsigned	lwr	
5(101)	store byte	store half	swl	store word			swr	
6(110)	load linked word	lwc1						
7(111)	store cond. word	swc1						
op(31:26)=010000 (TLB), rs(25:21)								
23-21	0(000)	1(001)	2(010)	3(011)	4(100)	5(101)	6(110)	7(111)
25-24								
0(00)	mfc0		cfc0		mtc0		ctc0	
1(01)								
2(10)								
3(11)								



op(31:26)=000000 (R-format), funct(5:0)								
2-0 5-3	0(000)	1(001)	2(010)	3(011)	4(100)	5(101)	6(110)	7(111)
0(000)	shift left logical		shift right logical	sra	sllv		srlv	srav
1(001)	jump register	jalr			syscall	break		
2(010)	mfhi	mthi	mflo	mtlo				
3(011)	mult	multu	div	divu				
4(100)	add	addu	subtract	subu	and	or	xor	not or (nor)
5(101)			set l.t.	set l.t. unsigned				
6(110)								
7(111)								

